HumanCLAW: Vision-Language Models Lose Track of Their Own Bodies

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Meta-led researchers built HumanCLAW to test whether vision-language models can control a body, not just describe a camera view. Every 0.5 seconds, a frozen model sees through a simulated human's eyes and chooses one movement, such as walking, turning, or sitting. A motion generator executes the command while a simulator applies gravity and collisions without letting balance or motor errors decide the result. HumanCLAW-Bench contains 1,218

find, navigate, and interact tasks across 41 indoor scenes. Nine models attempted every task. The best, Gemini 3.1, recognized the target in 64.9% of episodes, reached it in 42.4%, and completed the interaction in 16.8%.

Seeing the object was not enough. Across nine models, agents found the target in 5,473 runs, then failed to reach and stop beside it in 68%. Nearly two-thirds of those failures came from losing track of the body's location, while body awareness contributed to 81% of interaction failures. A home robot with this weakness might identify a chair, then jam its foot against a table or sit in empty space. HumanCLAW isolates those decisions because every model uses the same motion system. The evidence is limited to simulation, and the repository says the code, weights, simulator, and benchmark data are not yet released.

HumanCLAW exposes a gap that object-recognition scores miss. A model can understand a room and still fail once a body moves through it. Embodied systems need to update their estimate of the body's position after every action, not only improve visual recognition.

Read More: RynnBrain combines perception, spatial reasoning, and action planning in one open model for robots.

Sources:

- HumanCLAW paper (arXiv)
- HumanCLAW project page and leaderboard
- HumanCLAW public repository and release plan

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